

EE2016: Microprocessors

Experiment 1: Computations using Atmel Atmega8 AVR through Assembly Program Emulation

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Aim:

To implement basic arithmetic and logical manipulation programs using Atmel Atmega8 microcontroller in assembly program emulation, including addition, multiplication and comparison.

Problems:

Problem 1 (Common 8-bit mathematical operation): Given two 8-bit binary words (byte), compute the sum and store it in a register.

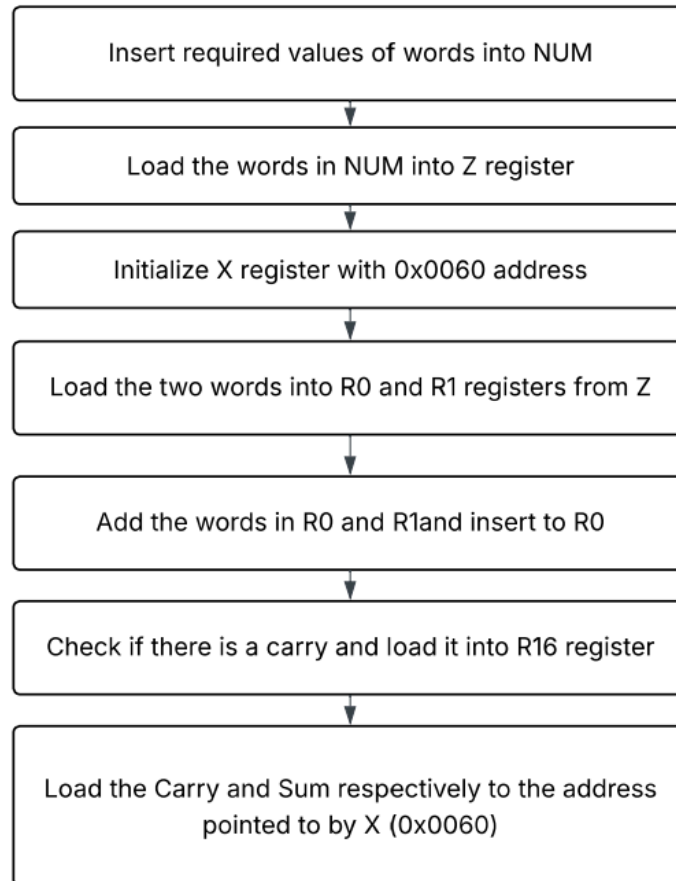
Problem 2 (16-bit addition using a 8-bit processor): Given two 16-bit binary words (byte), compute the sum and store it in a register.

Problem 3 (Multiplication of two 8-bit numbers): Given two 8-bit binary words (byte), compute the product and store it in a register.

Problem 4 (Largest of numbers given): Given a finite set of 16-bit binary words (5), identify the largest in the given set.

Problem 1:

Flow Chart:



Code:

```
.CSEG
LDI ZL,LOW(NUM<<1)
LDI ZH,HIGH(NUM<<1)
LDI XL,0x60
LDI XH,0x00
LDI R16,00
LPM R0,Z+
LPM R1,Z
ADD R0,R1
BRCC abc
LDI R16,0x01
abc: ST X+,R16
ST X,R0
NOP
NUM: .db 0xD3,0x5F
```

Output:

The screenshot displays the AVR Studio IDE interface during a simulation. The main window shows the assembly code for 'main.asm', with the instruction 'ADD R0,R1' highlighted. The 'Processor Status' window on the right provides a summary of the processor's state:

Name	Value
Program Counter	0x00000007
Stack Pointer	0x0000
X Register	0x0060
Y Register	0x0000
Z Register	0x001B
Status Register	T T H S V N Z C
Cycle Counter	11
Frequency	1.000 MHz
Stop Watch	11.00 µs

Below the processor status, the 'Registers' window shows the values of R0 through R7:

Register	Value
R00	0xD3
R01	0x5F
R02	0x00
R03	0x00
R04	0x00
R05	0x00
R06	0x00
R07	0x00

The 'Memory' window at the bottom right shows the contents of 'prog FLASH' starting at address 0x0060, displaying a series of zero-filled bytes.

FileEditViewVAssistXASFProjectBuildDebugToolsWindowHelp

Debug

Debug Browser

HexATmega8Simul

Experiment_1.1.asmmain.asm

```
.CSEG
LDI ZL,LOW(NUM<<1)
LDI ZH,HIGH(NUM<<1)
LDI XL,0x60
LDI XH,0x00
LDI R16,00
LPM R0,Z+
LPM R1,Z
ADD R0,R1
BRCC abc
LDI R16,0x01
abc: ST X+,R16
ST X,R0
NOP
NUM: .db 0xD3,0x5F
```

100 %

Autos

LocalsWatch 1Watch 2

Processor Status

Name	Value
R08	0x00
R09	0x00
R10	0x00
R11	0x00
R12	0x00
R13	0x00
R14	0x00
R15	0x00
R16	0x01
R17	0x00
R18	0x00
R19	0x00
R20	0x00
R21	0x00
R22	0x00
R23	0x00
R24	0x00
R25	0x00

Solution Explorer

Properties

Memory 4

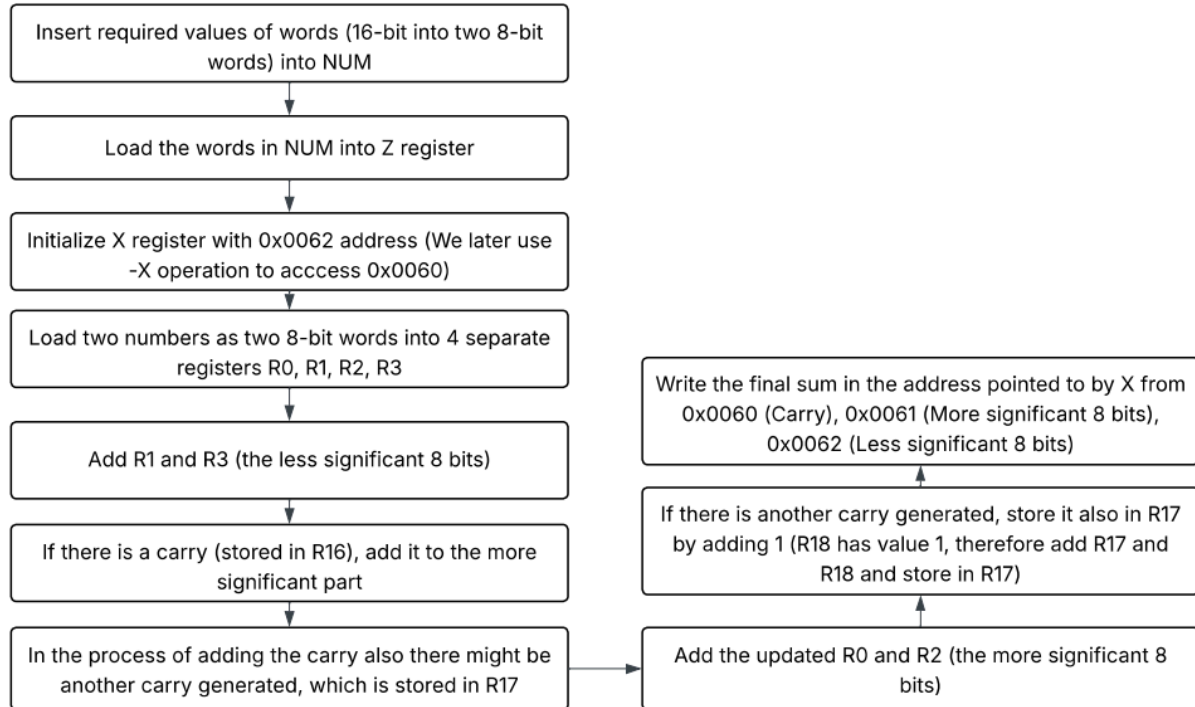
Memory: prog FLASHAddress: 0x0060,data

data 0x0060	01 32 00 00 00 00 00 00 00 00 00 00 00 00 00 00	.2.....
data 0x0071	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x0082	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x0093	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x00A4	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x00B5	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x00C6	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x00D7	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x00E8	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	

Call StackBreakpointsCommand WindowImmediate WindowOutputMemory 4

Problem 2:

Flow Chart:



Code:

```
.CSEG
LDI ZL,LOW(NUM<<1)
LDI ZH,HIGH(NUM<<1)
LDI XL,0x62
LDI XH,0x00
LDI R16, 0x00
LDI R17, 0x00
LDI R18, 0x01

LPM R0,Z+
LPM R1,Z+
LPM R2,Z+
LPM R3,Z

ADD R1,R3
BRCC no_carry_1
LDI R16,0x01
no_carry_1:
ST X, R1

ADD R0, R16
BRCC no_carry_2
LDI R17,0x01
no_carry_2:

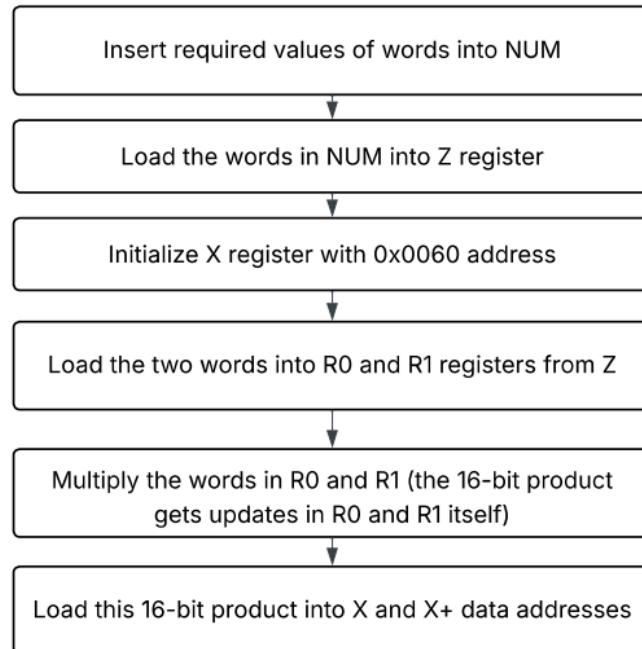
ADD R0, R2
BRCC no_carry_3
ADD R17, R18
no_carry_3:
ST -X, R0

ST -X, R17

NOP
NUM: .db 0x1A, 0x3F, 0x0B, 0x25
```


Problem 3:

Flow Chart:

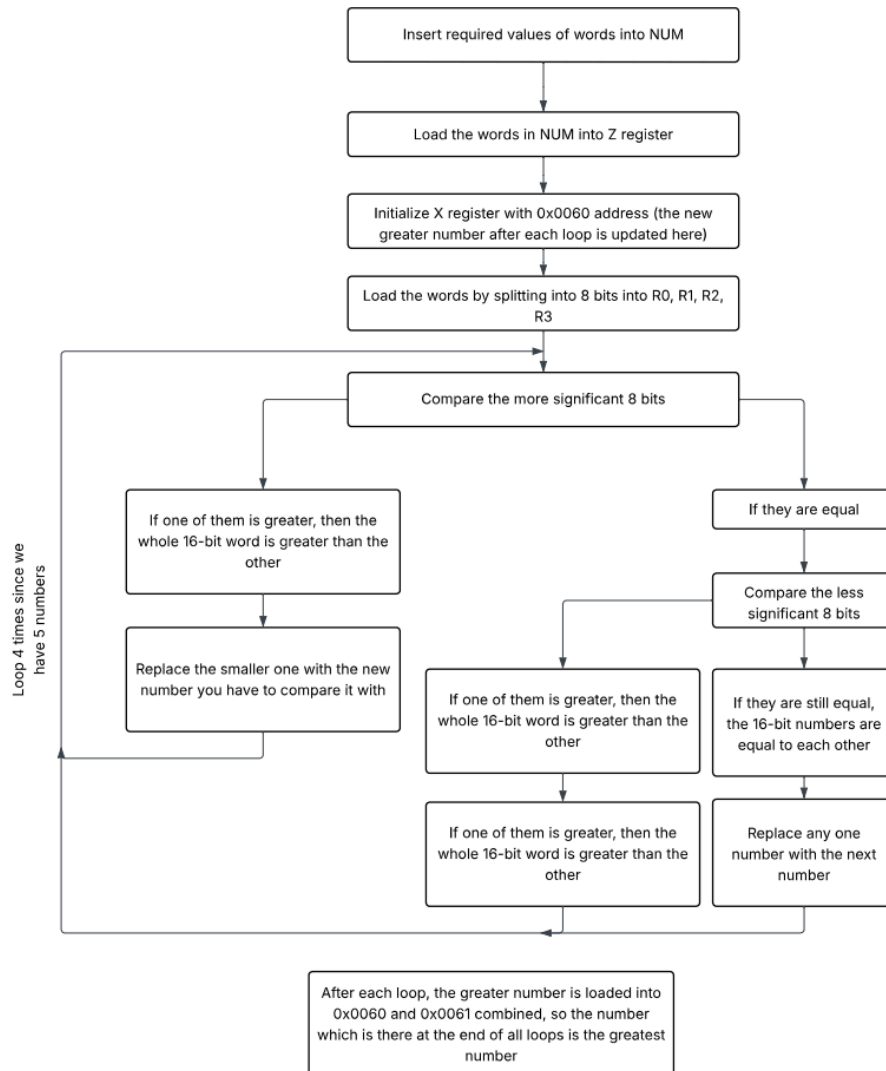


Code:

```
.CSEG
LDI ZL,LOW(NUM<<1)
LDI ZH,HIGH(NUM<<1)
LDI XL,0x60
LDI XH,0x00
LDI R16,00
LPM R0,Z+
LPM R1,Z
MUL R0,R1
ST X+,R1
ST X,R0
NOP
NUM: .db 0xD3,0x5F
```


Problem 4:

Flow Chart:



Code:

```
.CSEG
LDI ZL, LOW(NUM<<1)
LDI ZH, HIGH(NUM<<1)

LDI R20, 0x04
LPM R0,Z+
LPM R1,Z+
LPM R2,Z+
LPM R3,Z+

loop_start:
LDI XL, 0x60
LDI XH, 0x00

CP R0, R2
BREQ equal
BRCS lesser
rjmp greater

equal:
CP R1, R3
BREQ inner_equal
BRCS inner_lesser
rjmp inner_greater
```

inner_equal:

LPM R0, Z+

LPM R1, Z+

ST X+, R2

ST X, R3

rjmp done

inner_lesser:

LPM R0, Z+

LPM R1, Z+

ST X+, R2

ST X, R3

rjmp done

inner_greater:

LPM R2, Z+

LPM R3, Z+

ST X+, R0

ST X, R1

lesser:

LPM R0, Z+

LPM R1, Z+

ST X+, R2

ST X, R3

rjmp done

greater:

LPM R2, Z+

LPM R3, Z+

ST X+, R0

ST X, R1

done:

DEC R20

BRNE loop_start

NOP

NUM: .db 0x00, 0x01, 0x00, 0x34, 0x7F, 0xFF, 0xAB, 0xCD, 0xFF, 0xFF, 0x00, 0x00
; 1, 52, 32767, 43981, 65535, 0

Output:

The screenshot shows the AVR Studio IDE with the assembly code for 'main.asm' open. The code includes three subroutines: 'inner_equal', 'inner_lesser', and 'inner_greater'. The 'inner_lesser' subroutine is currently selected, showing instructions like 'LPM R0, Z+', 'LPM R1, Z+', 'ST X+, R2', 'ST X, R3', and 'rjmp done'. The 'Processor Status' window on the right displays the following values:

Name	Value
Program Counter	0x00000017
Stack Pointer	0x0000
X Register	0x0060
Y Register	0x0000
Z Register	0x005B
Status Register	01111010
Cycle Counter	27
Frequency	1.000 MHz
Stop Watch	27.00 µs

The 'Registers' window shows the following values:

Register	Value
R00	0x7F
R01	0x01
R02	0x00
R03	0x34
R04	0x00
R05	0x00
R06	0x00
R07	0x00

The 'Memory' window shows the 'prog FLASH' memory at address 0x0060, with data values ranging from 0x00 to 0xFF.

The screenshot shows the AVR Studio IDE with the assembly code for 'main.asm' open. The code includes instructions like 'LDI ZH, HIGH(NUM<<1)', 'LDI R20, 0x04', 'LPM R0, Z+', 'LPM R1, Z+', 'LPM R2, Z+', 'LPM R3, Z+', 'loop_start:', 'LDI XL, 0x60', 'LDI XH, 0x00', 'CP R0, R2', 'BREQ equal', 'BRCS lesser', 'rjmp greater', 'equal:', and 'CP R1, R3'. The 'Processor Status' window on the right displays the following values:

Name	Value
Program Counter	0x00000007
Stack Pointer	0x0000
X Register	0x0061
Y Register	0x0000
Z Register	0x005C
Status Register	01111010
Cycle Counter	39
Frequency	1.000 MHz
Stop Watch	39.00 µs

The 'Registers' window shows the following values:

Register	Value
R00	0x7F
R01	0xFF
R02	0x00
R03	0x34
R04	0x00
R05	0x00
R06	0x00
R07	0x00

The 'Memory' window shows the 'prog FLASH' memory at address 0x0060, with data values ranging from 0x00 to 0xFF.

File Edit View VAssistX ASF Project Build Debug Tools Window Help

Experiment_1.4.asm Experiment_1.3.asm Experiment_1.1.asm main.asm

```
lesser:
LPM R0, Z+
LPM R1, Z+
ST X+, R2
ST X, R3
rjmp done

greater:
LPM R2, Z+
LPM R3, Z+
ST X+, R0
ST X, R1

done:
DEC R20
BRNE loop_start

NOP
```

100 %

Autos

Name	Value	Type
------	-------	------

Processor Status

Name	Value
Program Counter	0x00000028
Stack Pointer	0x0000
X Register	0x0061
Y Register	0x0000
Z Register	0x005E
Status Register	11111111
Cycle Counter	56
Frequency	1.000 MHz
Stop Watch	56.00 µs

Registers

R00	0x7F
R01	0xFF
R02	0xAB
R03	0xCD
R04	0x00
R05	0x00
R06	0x00
R07	0x00

Memory 4

Memory: prog FLASH Address: 0x0060,data

data 0x0060	7f ff 00 00 00 00 00 00 00 00 00 00 00 00 00 00	.j.....
data 0x0071	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x0082	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x0093	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x00A4	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x00B5	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x00C6	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x00D7	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x00E8	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x00F9	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	

Autos Locals Watch 1 Watch 2

Call Stack Breakpoints Command Window Immediate Window Output Memory 4

File Edit View VAssistX ASF Project Build Debug Tools Window Help

Experiment_1.4.asm Experiment_1.3.asm Experiment_1.1.asm main.asm

```
LPM R1, Z+
ST X+, R2
ST X, R3
rjmp done

greater:
LPM R2, Z+
LPM R3, Z+
ST X+, R0
ST X, R1

done:
DEC R20
BRNE loop_start

NOP
NUM: .db 0x00, 0x01, 0x00, 0x34, 0x7F, 0xFF, 0xAB, 0xCD, 0xFF, 0xFF, 0x00, 0x00
; 1, 52, 32767, 43981, 65535, 0
```

100 %

Autos

Name	Value	Type
------	-------	------

Processor Status

Name	Value
Program Counter	0x00000028
Stack Pointer	0x0000
X Register	0x0061
Y Register	0x0000
Z Register	0x0062
Status Register	11111111
Cycle Counter	100
Frequency	1.000 MHz
Stop Watch	100.00 µs

Registers

R00	0xFF
R01	0xFF
R02	0x00
R03	0x00
R04	0x00
R05	0x00
R06	0x00
R07	0x00

Memory 4

Memory: prog FLASH Address: 0x0060,data

data 0x0060	ff ff 00 00 00 00 00 00 00 00 00 00 00 00 00 00	yy.....
data 0x0071	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x0082	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x0093	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x00A4	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x00B5	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x00C6	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x00D7	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x00E8	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
data 0x00F9	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	

Autos Locals Watch 1 Watch 2

Call Stack Breakpoints Command Window Immediate Window Output Memory 4